30 CTE Questions and Answers

Schema for Questions

```
CREATE TABLE employees (
  id INT PRIMARY KEY,
  name VARCHAR(100),
  department VARCHAR(100),
  manager_id INT,
  salary DECIMAL(10,2),
  hire_date DATE
);
```

```
INSERT INTO employees (id, name, department, manager_id, salary, hire_date) VALUES
(1, 'Alice', 'HR', NULL, 70000, '2015-06-23'),
(2, 'Bob', 'IT', 1, 90000, '2016-09-17'),
(3, 'Charlie', 'Finance', 1, 80000, '2017-02-01'),
(4, 'David', 'IT', 2, 75000, '2018-07-11'),
(5, 'Eve', 'Finance', 3, 72000, '2019-04-30');
```

Recursive CTEs

1. How do you retrieve employees and their hierarchical managers recursively?

```
WITH EmployeeHierarchy AS (
```

```sql
  SELECT id, name, manager_id, 1 AS level
  FROM employees WHERE manager_id IS NULL
  UNION ALL
  SELECT e.id, e.name, e.manager_id, eh.level + 1
  FROM employees e JOIN EmployeeHierarchy eh ON e.manager_id = eh.id
)
SELECT * FROM EmployeeHierarchy ORDER BY level;
```

2. How do you calculate the number of levels in an organizational hierarchy?

```sql
WITH Hierarchy AS (
  SELECT id, name, manager_id, 1 AS level FROM employees WHERE manager_id IS NULL
  UNION ALL
  SELECT e.id, e.name, e.manager_id, h.level + 1
  FROM employees e JOIN Hierarchy h ON e.manager_id = h.id
)
SELECT MAX(level) AS max_levels FROM Hierarchy;
```

3. How can you calculate cumulative salary within each department?

```sql
WITH RunningSalary AS (
  SELECT department, name, salary,
      SUM(salary) OVER (PARTITION BY department ORDER BY hire_date) AS running_total
  FROM employees
)
SELECT * FROM RunningSalary;
```

4. How do you find the most recent hire in each department?

```sql
WITH LatestHire AS (
  SELECT department, name, hire_date,
      RANK() OVER (PARTITION BY department ORDER BY hire_date DESC) AS rnk
  FROM employees
)
SELECT * FROM LatestHire WHERE rnk = 1;
```

5. How can you identify employees without direct reports?

```sql
WITH EmployeeReport AS (
  SELECT e1.id, e1.name, COUNT(e2.id) AS report_count
  FROM employees e1 LEFT JOIN employees e2 ON e1.id = e2.manager_id
  GROUP BY e1.id, e1.name
)
SELECT * FROM EmployeeReport WHERE report_count = 0;
```

Ranking & Aggregation

6. How do you rank employees based on salary within each department?

```sql
WITH SalaryRanking AS (
  SELECT id, name, department, salary,
      RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS rank
  FROM employees
)
SELECT * FROM SalaryRanking WHERE rank <= 3;
```

7. How do you find employees who are earning below the department's average salary?

```
WITH DeptAvg AS (
  SELECT department, AVG(salary) AS avg_salary FROM employees GROUP BY department
)
SELECT e.* FROM employees e
JOIN DeptAvg d ON e.department = d.department
WHERE e.salary < d.avg_salary;
```

8. How do you find the median salary per department using a CTE?

```
WITH SalaryRanks AS (
  SELECT department, salary,
      ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary) AS rn,
      COUNT(*) OVER (PARTITION BY department) AS total
  FROM employees
)
SELECT department, AVG(salary) AS median_salary
FROM SalaryRanks WHERE rn IN ((total + 1) / 2, (total + 2) / 2) GROUP BY department;
```

9. How do you identify employees hired in the same year as their manager?

```
WITH HireYears AS (
  SELECT e1.name AS emp_name, e1.hire_date AS emp_hire,
      e2.name AS mgr_name, e2.hire_date AS mgr_hire
  FROM employees e1 JOIN employees e2 ON e1.manager_id = e2.id
)
```

```sql
SELECT * FROM HireYears WHERE EXTRACT(YEAR FROM emp_hire) = EXTRACT(YEAR FROM mgr_hire);
```

10. How do you calculate rolling 3-month average salary per department?

```sql
WITH RollingSalary AS (
  SELECT department, name, hire_date, salary,
      AVG(salary) OVER (PARTITION BY department ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) AS rolling_avg
  FROM employees
)
SELECT * FROM RollingSalary;
```

---

11. How do you identify departments where salaries are increasing over time?

```sql
WITH SalaryTrend AS (
  SELECT department, name, salary, hire_date,
      LAG(salary) OVER (PARTITION BY department ORDER BY hire_date) AS prev_salary
  FROM employees
)
SELECT * FROM SalaryTrend WHERE salary > prev_salary;
```

---

12. How do you find employees hired after their manager?

```sql
WITH ManagerHires AS (
  SELECT e1.id AS emp_id, e1.name AS emp_name, e1.hire_date AS emp_hire,
```

```sql
        e2.name AS mgr_name, e2.hire_date AS mgr_hire
    FROM employees e1 JOIN employees e2 ON e1.manager_id = e2.id
)
SELECT * FROM ManagerHires WHERE emp_hire > mgr_hire;
```

---

13. How do you determine the highest and lowest-paid employees using a CTE?

```sql
WITH SalaryExtremes AS (
    SELECT name, salary,
        FIRST_VALUE(name) OVER (ORDER BY salary DESC) AS highest_paid,
        FIRST_VALUE(name) OVER (ORDER BY salary ASC) AS lowest_paid
    FROM employees
)
SELECT DISTINCT highest_paid, lowest_paid FROM SalaryExtremes;
```

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14. How can you use a CTE to generate Fibonacci numbers?

```sql
WITH RECURSIVE Fibonacci (n, fib) AS (
    SELECT 1, 0 UNION ALL
    SELECT n + 1, fib + (SELECT fib FROM Fibonacci WHERE n = Fibonacci.n - 1) FROM Fibonacci WHERE n < 10
)
SELECT * FROM Fibonacci;
```

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15. How do you pivot department-wise employee counts using a CTE?

```
WITH DeptCount AS (
  SELECT department, COUNT(*) AS emp_count FROM employees GROUP BY department
)
SELECT * FROM DeptCount;
```

---

16. How do you calculate the difference in salary between an employee and their manager?

```
WITH SalaryComparison AS (
  SELECT e1.name AS emp_name, e1.salary AS emp_salary, e2.name AS mgr_name, e2.salary AS mgr_salary,
      (e1.salary - e2.salary) AS salary_diff
  FROM employees e1 JOIN employees e2 ON e1.manager_id = e2.id
)
SELECT * FROM SalaryComparison;
```

---

17. How can you find the longest-tenured employees using a CTE?

```
WITH Tenure AS (
  SELECT name, hire_date,
      RANK() OVER (ORDER BY hire_date ASC) AS tenure_rank
```

```
  FROM employees
)
SELECT * FROM Tenure WHERE tenure_rank = 1;
```

---

18. How can you create a CTE that identifies salary gaps between employees?

```
WITH SalaryGaps AS (
  SELECT name, salary,
      LAG(salary) OVER (ORDER BY salary) AS prev_salary,
      salary - LAG(salary) OVER (ORDER BY salary) AS salary_gap
  FROM employees
)
SELECT * FROM SalaryGaps;
```

---

19. How do you filter only employees who received a salary increase over time?

```
WITH SalaryChanges AS (
  SELECT name, salary, hire_date,
      LAG(salary) OVER (PARTITION BY name ORDER BY hire_date) AS prev_salary
  FROM employees
)
SELECT * FROM SalaryChanges WHERE salary > prev_salary;
```

---

20. How do you rank employees within their department based on hire date using a CTE?

```
WITH HireRanking AS (
  SELECT department, name, hire_date,
      RANK() OVER (PARTITION BY department ORDER BY hire_date) AS hire_rank
  FROM employees
)
SELECT * FROM HireRanking;
```

---

21. How do you use a CTE to determine the cumulative headcount per year?

```
WITH YearlyHires AS (
  SELECT EXTRACT(YEAR FROM hire_date) AS hire_year, COUNT(*) AS yearly_hires
  FROM employees GROUP BY hire_year
)
SELECT hire_year, SUM(yearly_hires) OVER (ORDER BY hire_year) AS cumulative_hires
FROM YearlyHires;
```

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22. How do you generate a sequence of numbers using a recursive CTE?

```
WITH RECURSIVE NumberSequence AS (
  SELECT 1 AS num
  UNION ALL
```

```
  SELECT num + 1 FROM NumberSequence WHERE num < 10
)
SELECT * FROM NumberSequence;
```

---

23. How do you use a CTE to remove duplicate salaries while keeping the latest employee?

```
WITH RankedEmployees AS (
  SELECT *, ROW_NUMBER() OVER (PARTITION BY salary ORDER BY hire_date DESC) AS rn
  FROM employees
)
SELECT * FROM RankedEmployees WHERE rn = 1;
```

---

24. How do you determine which employees report directly or indirectly to a given manager?

```
WITH RECURSIVE EmployeeHierarchy AS (
  SELECT id, name, manager_id FROM employees WHERE manager_id = 1
  UNION ALL
  SELECT e.id, e.name, e.manager_id FROM employees e JOIN EmployeeHierarchy eh ON e.manager_id = eh.id
)
SELECT * FROM EmployeeHierarchy;
```

---

25. How do you find the first and last employee hired in each department?

```
WITH HireOrder AS (
  SELECT department, name, hire_date,
      FIRST_VALUE(name) OVER (PARTITION BY department ORDER BY hire_date) AS first_hire,
      LAST_VALUE(name) OVER (PARTITION BY department ORDER BY hire_date ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING) AS last_hire
  FROM employees
)
SELECT DISTINCT department, first_hire, last_hire FROM HireOrder;
```

---

26. How do you calculate the percentage of total salary each employee contributes within a department?

```
WITH SalaryShare AS (
  SELECT department, name, salary,
      salary * 100.0 / SUM(salary) OVER (PARTITION BY department) AS percentage
  FROM employees
)
SELECT * FROM SalaryShare;
```

---

27. How do you identify departments where salary expenses exceed a given threshold?

```
WITH DeptSalaries AS (
  SELECT department, SUM(salary) AS total_salary FROM employees GROUP BY department
)
SELECT * FROM DeptSalaries WHERE total_salary > 150000;
```

---

28. How do you identify employees earning within a certain percentile using a CTE?

```
WITH SalaryPercentile AS (
  SELECT name, salary,
      NTILE(4) OVER (ORDER BY salary DESC) AS salary_quartile
  FROM employees
)
SELECT * FROM SalaryPercentile WHERE salary_quartile = 1;
```

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29. How do you generate a list of missing salary brackets for employees?

```
WITH RecursiveBrackets AS (
  SELECT 50000 AS salary_bracket
  UNION ALL
  SELECT salary_bracket + 5000 FROM RecursiveBrackets WHERE salary_bracket < 150000
)
```

```sql
SELECT salary_bracket FROM RecursiveBrackets WHERE salary_bracket NOT IN
(SELECT DISTINCT salary FROM employees);
```

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30. How do you use a CTE to get the top 3 highest-paid employees in each department?

```sql
WITH TopSalaries AS (
  SELECT name, department, salary,
      DENSE_RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS rnk
  FROM employees
)
SELECT * FROM TopSalaries WHERE rnk <= 3;
```